

ECE456 Network Centric Programming – Socket Programming

1. (6 points) Fill in the missing line to send a short message.

```
1 #include <sys/socket.h>
2
3 int main() {
4
5     int sock;
6
7     const char* msg = "hello";
8
9     _____(sock, msg, 5, 0);
10 }
```

2. (6 points) What condition indicates an error from `recv()`?

```
1 int n = recv(sock, buf, 1000, 0);
```

Circle the correct answer.

- $n < 0$
- $n < 1000$
- $n > 0$

3. (6 points) What does a return value of `**0**` from `recv()` indicate?

Short answer.

4. (8 points) What is wrong with the following code?

(single phrase)

```
1 int n = recv(sock, buf, 1024, 0);
2
3 if(n < 1024) {
4     printf("error\n");
5 }
```

Answer:

5. (10 points) Consider the following client code.

```
1 const char* msg = "HELLO";
2
3 send(sock, msg, 5, 0);
```

Is it guaranteed that all 5 bytes are sent?

- Yes
- No

If no, what must the program do?

- retry in a loop
- reopen the socket
- call fork()

6. (10 points) Fill in the missing code so the program sends the entire buffer.

```
1 int sent = 0;
2
3 while(sent < len) {
4
5     int n = send(sock,
6                 buf + sent,
7                 len - sent,
8                 0);
9
10    if(n < 0)
11        return -1;
12
13    -----;
14 }
```

7. (10 points) Fill in the missing code so the program receives a full message.

```
1 int received = 0;
2
3 while(received < len) {
4
5     int n = recv(sock,
6                 buf + received,
7                 len - received,
8                 0);
9
10    if(n < 0)
11        return -1;
12
13    -----;
14 }
```

8. (8 points) What does the following IP address represent?

127.0.0.1

Short answer.

9. (8 points) Which protocol guarantees reliable delivery?

- UDP
- TCP

10. (10 points) A TCP client sends a message.

The client later receives an ****ACK**** packet.

Does this guarantee that the server application has processed the message?

- Yes
- No

11. (10 points) What layer generates the ACK packet?

- Application
- TCP
- IP

12. (12 points) Consider the following server code.

```
1 int client = accept(server_fd, NULL, NULL);
2
3 char buf[100];
4
5 recv(client, buf, 100, 0);
6
7 printf("%s\n", buf);
```

Is it guaranteed that the full message was received?

- Yes
- No

Brief reason:

13. (12 points) What is wrong with the following code?

(single phrase)

```
1 char buf[100];
2
3 int n = recv(sock, buf, 100, 0);
4
5 printf("%s\n", buf);
```

Answer:

14. (15 points) Write a loop that reads from a socket until the connection closes.

Use:

- recv
- break when recv returns 0